

MATH 2400 - Week 2 Notes

1 Lines

Let $P(x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$ and let $\vec{v} = \langle a, b, c \rangle$ be a vector. Let $\vec{r}_0 = \overrightarrow{OP}$ and $\vec{r} = \langle x, y, z \rangle$. Then the line through P in the direction $\vec{v}$ is given by

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

This is called the **vector equation** of the line. The value t is called a **parameter** and $-\infty \leq t \leq \infty$.

Looking at the components, we get the **parametric equations** for the line:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

If we isolate for t and set equal, we get the **symmetric equations** of the line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Given two lines with vector equations $\vec{r} = \vec{r}_1 + t\vec{v}_1$ and $\vec{r} = \vec{r}_2 + t\vec{v}_2$, then the lines are parallel if and only if $\vec{v}_1 = k\vec{v}_2, k \neq 0$, that is, if the direction vectors are parallel.

In $\mathbb{R}^3$ lines that are not parallel and do not intersect are called **skew lines**.

2 Planes

A plane is uniquely determined by any two distinct non-skew lines (that is, they either intersect or are parallel), any one line and a point not on it, any two distinct non-parallel vectors, or any three non-collinear points.

Given any three **non-collinear** points, P, Q, R , in the plane, there is a unique plane that contains the points.

Let $\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \langle a, b, c \rangle$ and let $\vec{r}_0 = \overrightarrow{OP} = \langle x_0, y_0, z_0 \rangle$. Then $\vec{n}$ is the **normal vector** to the plane and we have

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$

This is called the **vector equation** of the plane.

If we multiply out the dot product, we get the **scalar equation** of the plane:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

If we gather all the constants together, we get the **linear equation** of the plane:

$$ax + by + cz + d = 0, \quad \text{with } d = -(ax_0 + by_0 + cz_0)$$

When working with actual numerical values, we will often move the constant d to the other side, in which case we have

$$ax + by + cz = -d$$

which gives

$$ax + by + cz = ax_0 + by_0 + cz_0$$

There is a certain elegance and symmetry to writing it this way.

Notice that the normal vector can be found from either the scalar or the linear equation by taking the coefficients of the x , y , and z terms as components (in that order). That is, given a scalar or linear equation as written above, then $\vec{n} = \langle a, b, c \rangle$.

Two planes are parallel if their normal vectors are parallel.

If two planes are not parallel, then they intersect in a line. If $\vec{n}_1$ and $\vec{n}_2$ are the two normal vectors of the plane, then the direction vector of the line of intersection is $\vec{v} = \vec{n}_1 \times \vec{n}_2$. This is saying that the line of intersection is orthogonal to both normal vectors.

The angle between the planes is the acute angle between the normal vectors. This is given by

$$\cos(\theta) = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

The distance from a point, $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by

$$D = |\text{comp}_{\vec{n}} \vec{b}| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Here, $\vec{b} = \overrightarrow{P_0P_1}$, where P_0 is any point in the plane and $\vec{n} = \langle a, b, c \rangle$ is the normal to the plane.

Example: Find the plane that contains $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

First, $\overrightarrow{PQ} = \langle 2, -4, 4 \rangle$ and $\overrightarrow{PR} = \langle 4, -1, -2 \rangle$. From this we find that

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{bmatrix} = \langle 12, 20, 14 \rangle$$

Then we have that the plane is given by $\vec{n} \cdot (\vec{r} - \overrightarrow{OP}) = 0$, which gives

$$\langle 12, 20, 14 \rangle \cdot \langle x - 1, y - 3, z - 2 \rangle = 0$$

Multiplying everything out and simplifying gives

$$6x + 10y + 7z = 50$$